

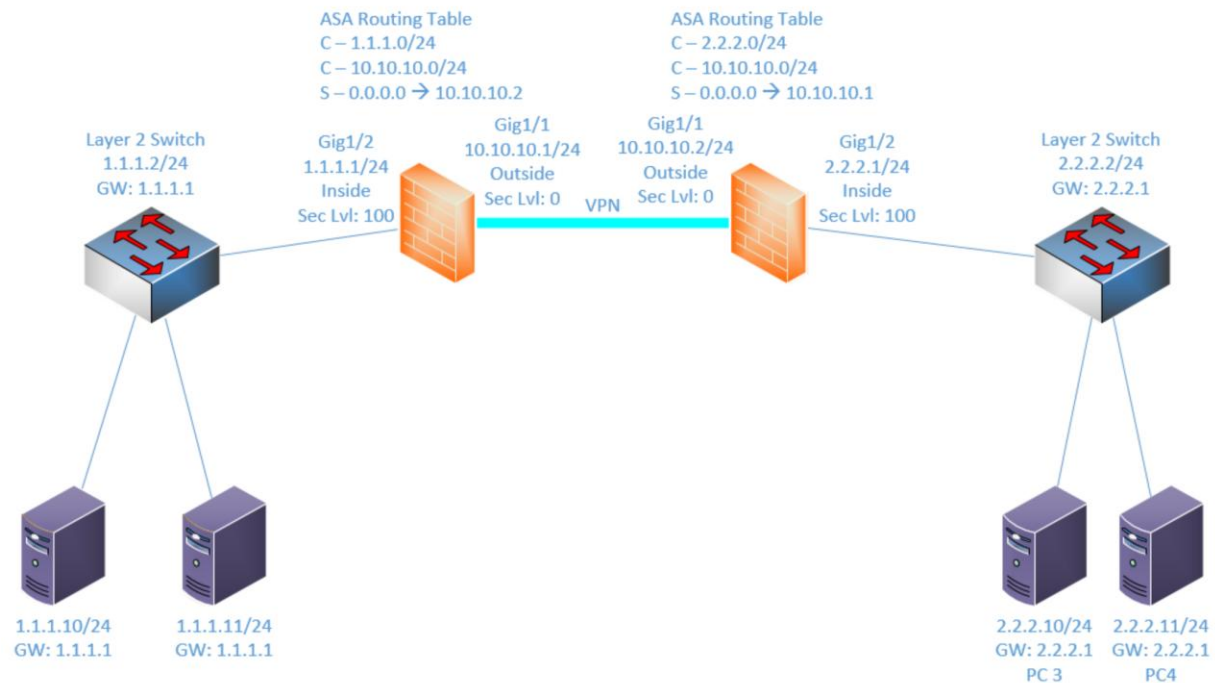
Lab 7

Implementing a Point to Point VPN

Objective:

The purpose of this lab is for the student to understand basic features of and implement a Point to Point VPN. Students will utilize the ASA GUI to explore and configure a Cisco firewall.

Diagram:



Components:

- 4 – PCs
- 2 – Cisco 3650 Layer 2 Switches
- 2 – Cisco 5516-X Firewalls

Procedure:

1. Connect all PC's together on the layer 2 switches and put their static IP addresses, subnet masks and gateways on them. Test connectivity by pinging each machine within the network.
2. Use the diagram above to label the firewalls IP addresses and ports below

	Name	IP	Interface	Security
Firewall 1	Outside	10.10.10.1/24	GigabitEthernet1/1	0
	Inside	1.1.1.1/24	GigabitEthernet1/2	100
Firewall 2	Outside	10.10.10.2/24	GigabitEthernet1/1	0
	Inside	2.2.2.1/24	GigabitEthernet1/2	100

3. Configure the IP addresses on the Firewalls' inside interfaces, set the name of the interface and the security level, and turn on the interfaces.
4. Enable http access to the firewall and allow the 1.1.1.0/24 network to access it from the inside interface on firewall 1.
5. Enable http access to the firewall and allow the 2.2.2.0/24 network to access it from the inside interface on firewall 2.
6. Ping each firewall's inside interfaces from a machine on the inside network and verify it is up and working.
7. Connect to the firewall's GUI through the web browser. (Unless the ASDM application is already on your desktop).
8. Select the configuration tab and under device setup select interfaces
9. Create the Interfaces for the outside network.

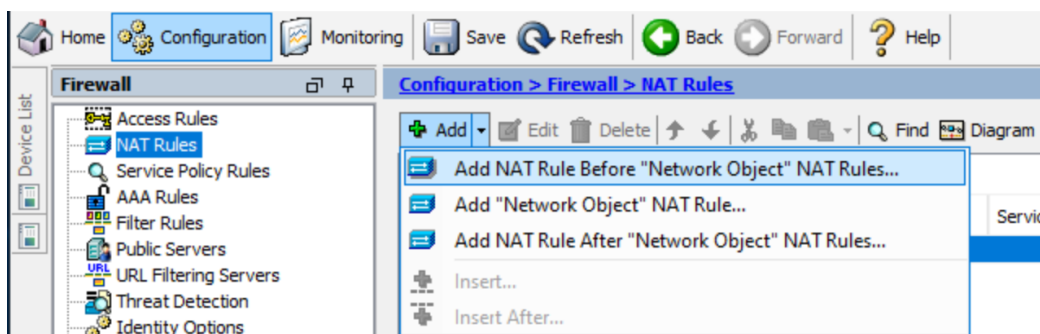
a. List the interface details below:

	Interface	IP	Security	Name
Firewall 1	GigabitEthernet1/1	10.10.10.1/24	0	Outside
Firewall 2	GigabitEthernet1/1	10.10.10.2/24	0	Outside

10. Add the default route and select the outside interface on both firewalls.
Refer to the diagram if necessary.

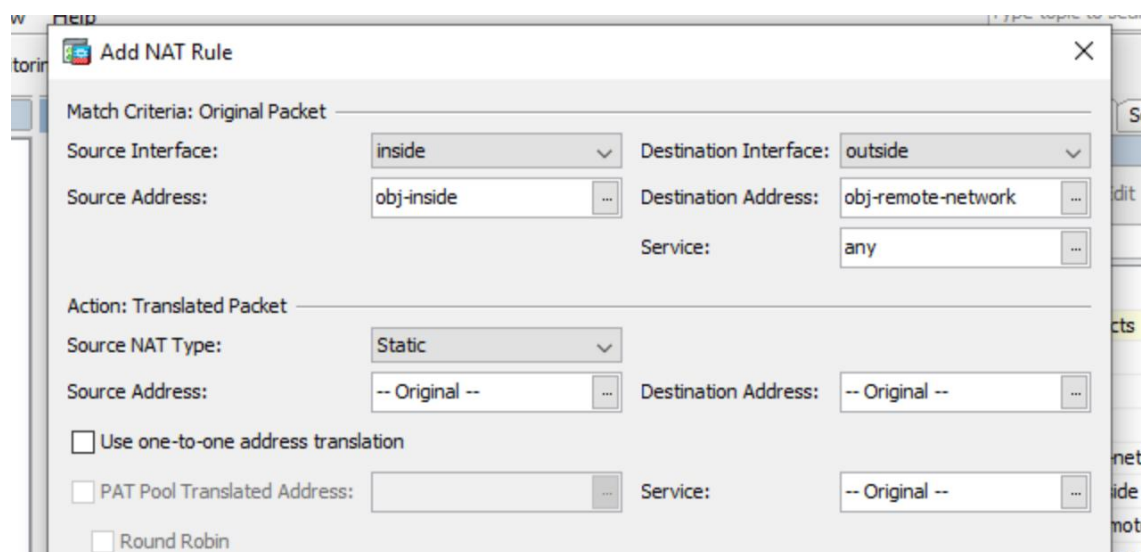
11. On Firewall1 Select Configuration, Firewall and click on NAT Rules.
Remember ASA NATs use objects in the address fields. Create objects
for any addresses you need for step 13.

12. Click the arrow next to Add and choose Add Nat rule before “Network
object” NAT rules.



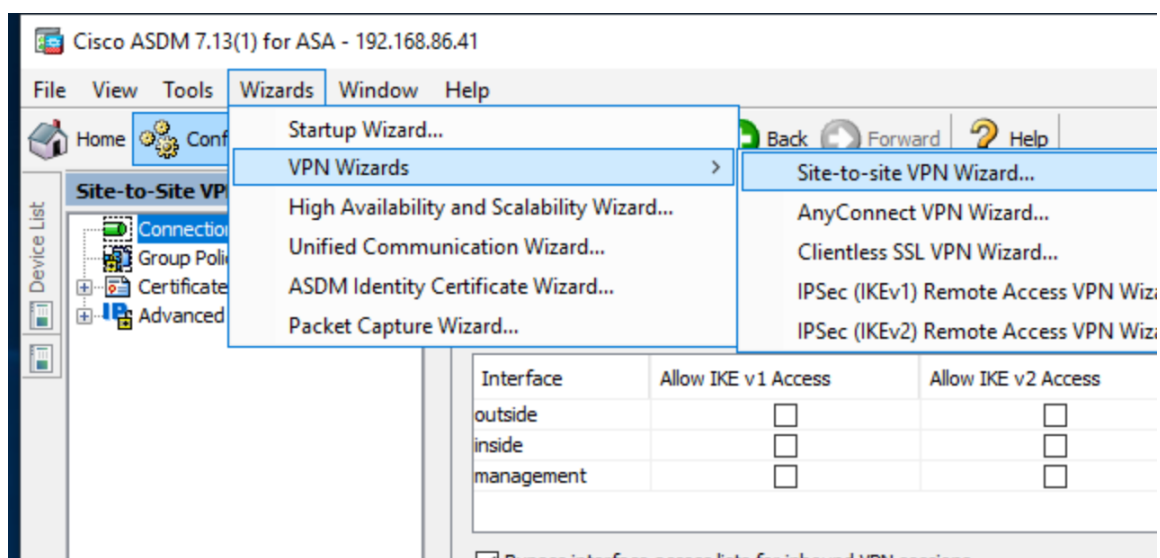
13. Create a NAT rule with the following:

- Source interface: inside
- Source address: 1.1.1.0/24
- Destination interface: outside
- Destination address: 2.2.2.0/24
- Translated source and destination original addresses.
- It should look similar to this:



14. Repeat steps 11-13 on Firewall 2, but make sure to reverse the source and destination addresses. The 2.2.2.0/24 network will be the source and 1.1.1.0/24 will be the destination.

15. On Firewall 1 select Wizards → VPN Wizards → Site to Site VPN Wizard

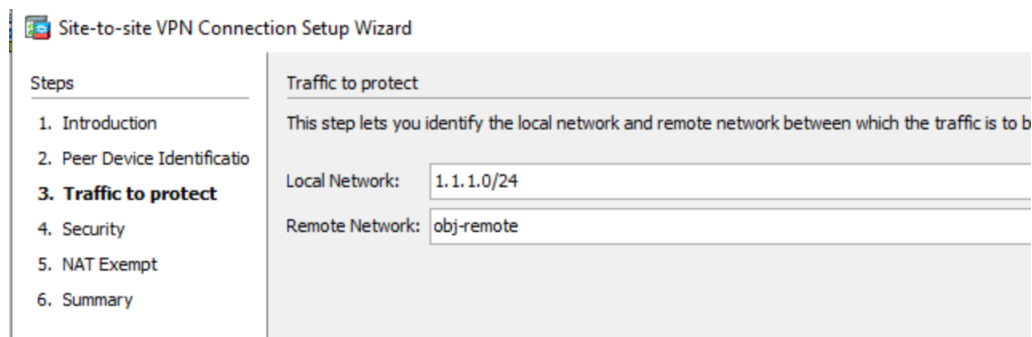


16. Select Next and then enter the IP address of Firewall 2's outside interface. Make sure the VPN access interface is outside. Click next when finished.



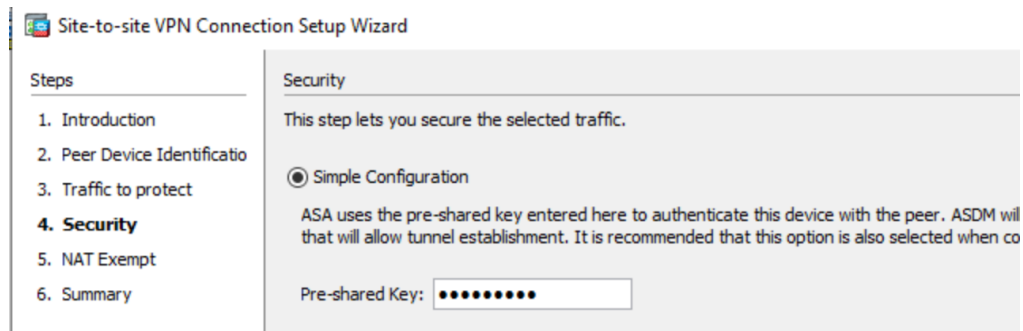
The screenshot shows the 'Peer Device Identification' step of the Site-to-site VPN Connection Setup Wizard. The left sidebar lists the steps: 1. Introduction, 2. Peer Device Identification (highlighted), 3. Traffic to protect, 4. Security, 5. NAT Exempt, and 6. Summary. The main panel has a title 'Peer Device Identification' and a description: 'This step lets you identify the peer VPN device by its IP address and the interface used to access'. It contains three input fields: 'Peer IP Address' with the value '10.10.10.2', 'Connection Name' which is empty, and 'VPN Access Interface' with the value 'outside'.

17. Select 1.1.1.0/24 as the local network and 2.2.2.0/24 as the remote network. Click next when finished.



The screenshot shows the 'Traffic to protect' step of the Site-to-site VPN Connection Setup Wizard. The left sidebar lists the steps: 1. Introduction, 2. Peer Device Identification, 3. Traffic to protect (highlighted), 4. Security, 5. NAT Exempt, and 6. Summary. The main panel has a title 'Traffic to protect' and a description: 'This step lets you identify the local network and remote network between which the traffic is to be protected'. It contains two input fields: 'Local Network' with the value '1.1.1.0/24' and 'Remote Network' with the value 'obj-remote'.

18. Set a preshared key of netcom123



The screenshot shows the 'Security' step of the Site-to-site VPN Connection Setup Wizard. The left sidebar lists the steps: 1. Introduction, 2. Peer Device Identification, 3. Traffic to protect, 4. Security (highlighted), 5. NAT Exempt, and 6. Summary. The main panel has a title 'Security' and a description: 'This step lets you secure the selected traffic.' It contains a radio button labeled 'Simple Configuration' which is selected. Below this is a text box for the 'Pre-shared Key' with the value 'netcom123'.

19. Click Next. Ignore the NAT Exempt part since we already set up NAT in the previous step.

20. Review the summary information. Notice all the additional information that is configured automatically for us. There are crypto policies and transform sets.

21. Click Finish

22. Can the 1.1.1.0/24 network ping the 2.2.2.0/24 Network? **No, it can't.**
a. Why? **The second Firewall hasn't been configured for VPN yet and is receiving traffic it cannot interpret**

23. On Firewall 2 select Wizards → VPN Wizards → Site to Site VPN Wizard

24. Follow steps 16-21 on Firewall 2. Remember the source and destinations are swapped since 2.2.2.0/24 is local and inside on this firewall.

25. Can the 1.1.1.0/24 network ping the 2.2.2.0/24 Network? **Yes**
a. Why? **Now both firewalls are configured for VPN**

26. Click on Monitoring and select the VPN Tab

27. Can you tell if the VPN tunnel is established? **Yes**

28. Does it show the peer IP addresses? **Yes, it shows 10.10.10.1 for firewall 2 and 10.10.10.2 for firewall 1**

29. Does it show the local and remote networks? **No**

30. Click on details.

- ## IPSEC

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FIREWALL 1

Cisco ASDM 7.16(1)150 for ASA - 1.1.1.1

File View Tools Wizards Window Help

Home Configuration Monitoring Save Refresh Back Forward Help

VPN

- VPN Statistics
 - Sessions
 - VPN Group Loads
 - Crypto Statistics
 - Compression Statistics
 - Encryption Statistics
 - Global IKE/IPsec Statistics
 - Protocol Statistics
 - VLAN Mapping Sessions
- Clientless SSL VPN
- Shared SSL VPN
- VPN Connection Graphs
- WSA Sessions

Interfaces

VPN

Routing

Properties

Logging

Monitoring > VPN > VPN Statistics > Sessions

Type	Active	Cumulative	Peak Concurrent	Inactive
Site-to-Site VPN	2	2	2	2
IKEv2 IPsec	1	1	1	1
IKEv1 IPsec	1	1	1	1

Filter By: IPsec Site-to-Site -- All Sessions -- Filter

Connection Profile IP Address	Protocol Encryption	Login Time Duration	Bytes Tx Bytes Rx	Cer Auth Int
10.10.10.2	IKEv2 IPsec	23:34:30 UTC Tue Oct 7 2025	420	
10.10.10.2	IKEv2: (1)AES256 IPsec: (...)	0h:00m:37s	420	
10.10.10.2	IKEv1 IPsec	23:34:33 UTC Tue Oct 7 2025	0	
10.10.10.2	IKEv1: (1)AES256 IPsec: (...)	0h:00m:34s	0	

Details Logout Ping

To sort VPN sessions, right-click on the above table and select Table Sort Order from popup menu.

Logout By: -- All Sessions -- Logout Sessions Refresh

Last Updated: 10/7/25 7:22:44 PM

Data Refreshed Successfully. <admin> 15 10/7/25 11:42:09 PM UTC

Session Details

Session Details

Connection Profile IP Address	Protocol Encryption	Login Time Duration	Bytes Tx Bytes Rx	Cer Auth Int	Cer Auth Left
10.10.10.2	IKEv2 IPsec	23:34:30 UTC Tue Oct 7 2025	420		
10.10.10.2	IKEv2: (1)AES256 IPsec: (1)AES256	0h:01m:44s	420		

Details ACL

ID	Type	Local Addr. / Subnet Mask / Protocol / Port Remote Addr. / Subnet Mask / Protocol / Port	Encryption	Other	Bytes Tx Bytes Rx
	IKEv2		AES-256	Tunnel ID: 2.1 Loc Auth Mode: preSharedKeys Rem Auth Mode: preSharedKeys UDP Source Port 500 UDP Destination Port 500 Authentication Mode: preSharedKeys UDP Source Port 500 UDP Destination Port 500 Hashing: SHA1 PRF: SHA1 Authentication Mode: preSharedKeys UDP Source Port 500 UDP Destination Port 500 IKE Negotiation Mode: none Hashing: SHA1 Diffie-Hellman Group: 5 Rekey Time Interval: 86400 Seconds Rekey Left(T): 86296 Seconds	
	IPsec	1.1.1.0/255.255.255.0/0/0 2.2.2.0/255.255.255.0/0/0	AES-256	Tunnel ID: 2.2 Hashing: SHA1 Encapsulation: Tunnel Rekey Time Interval: 28800 Seconds Rekey Left(T): 28696 Seconds Rekey Data Interval: 4608000 K-Bytes Rekey Left(D): 4608000 K-Bytes Idle Time Out: 30 Minutes Idle TO Left: 28 Minutes Packets Tx: 7 Packets Rx: 7	420 420

Refresh Close Help

Last Updated: 10/7/25 7:23:51 PM

FIREWALL 2

Cisco ASDM 7.16(1)150 for ASA - 2.2.1

File View Tools Wizards Window Help

Home Configuration Monitoring Save Refresh Back Forward Help

Device List Bookmarks Monitoring > VPN > VPN Statistics > Sessions

Device List

Ind: 1.1.1.1 2.2.2.1

VPN

VPN Statistics

VPN Group Loads

Crypto Statistics

Compression Statistics

Encryption Statistics

Global IKE/IPsec Statistics

Protocol Statistics

VLAN Mapping Sessions

Clientless SSL VPN

Shared SSL VPN

VPN Connection Graphs

WSA Sessions

Interfaces

VPN

Routing

Properties

Logging

Monitoring > VPN > VPN Statistics > Sessions

Filter By: IPsec Site-to-Site -- All Sessions -- Filter

Type	Active	Cumulative	Peak Concurrent	Inactive
Site-to-Site VPN	2	2	2	2
IKEv2 IPsec	1	1	1	1
IKEv1 IPsec	1	1	1	1

Connection Profile IP Address Protocol Encryption Login Time Duration Bytes Tx Bytes Rx Cer Auth Int Cer Auth Left

10.10.10.1	IKEv2 IPsec	23:35:01 UTC Tue Oct 7 2025	0h:00m:40s	420	420		
10.10.10.1	IKEv2: (1)AES256 IPsec: (...)	23:35:04 UTC Tue Oct 7 2025	0h:00m:37s	0	0		

To sort VPN sessions, right-click on the above table and select Table Sort Order from popup menu.

Logout By: -- All Sessions -- Logout Sessions Refresh

Last Updated: 10/7/25 7:22:47 PM

Data Refreshed Successfully.

Session Details

Session Details

Connection Profile	Protocol	Login Time	Bytes Tx	Cer Auth Int	Cer Auth Left
IP Address	Encryption	Duration	Bytes Rx		
10.10.10.1	IKEv2 IPsec	23:35:01 UTC Tue Oct 7 2025	420		
10.10.10.1	IKEv2: (1)AES256 IPsec: (...)	0h:05m:52s	420		

Details ACL

ID	Type	Local Addr. / Subnet Mask / Protocol / Port	Remote Addr. / Subnet Mask / Protocol / Port	Encryption	Other	Bytes Tx	Bytes Rx
	IKEv2			AES-256	Tunnel ID: 5.1 Loc Auth Mode: preSharedKeys Rem Auth Mode: preSharedKeys UDP Source Port 500 UDP Destination Port 500 Authentication Mode: preSharedKeys UDP Source Port 500 UDP Destination Port 500 Hashing: SHA1 PRF: SHA1 Authentication Mode: preSharedKeys UDP Source Port 500 UDP Destination Port 500 IKE Negotiation Mode: none Hashing: SHA1 Diffie-Hellman Group: 5 Rekey Time Interval: 86400 Seconds Rekey Left(T): 86047 Seconds		
	IPsec	2.2.2.0/255.255.255.0/0/0	1.1.1.0/255.255.255.0/0/0	AES-256	Tunnel ID: 5.2 Hashing: SHA1 Encapsulation: Tunnel Rekey Time Interval: 28800 Seconds Rekey Left(T): 28447 Seconds Rekey Data Interval: 4608000 K-Bytes Rekey Left(D): 4608000 K-Bytes Idle Time Out: 30 Minutes Idle TO Left: 24 Minutes Packets Tx: 7 Packets Rx: 7	420	420

Refresh Close Help

32. Draw the lab diagram below (PC's, router and switch connections) Label the IP addresses, subnet masks, and gateways used by each device.
Label the routing tables next to the switch, and firewalls.

